Natanael S. Alpay

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SUMMARY

Apply mathematical modeling and Python/PyTorch to implement and evaluate neural-network hedging strategies as a Mathematics Ph.D. candidate with 10+ publications and preprints. Communicate complex ideas through 20+ research talks, university teaching, and undergraduate mentorship. Seek quantitative research and applied machine-learning roles.

EDUCATION

Ph.D. Mathematics	University of California, Irvine	**Expected 2028**
M.S. Applied Mathematics	University of California, Irvine	**2025**
B.S. Mathematics	Chapman University	**2022**
B.S. Physics	Chapman University	**2022**
B.S. Computer Science	Chapman University	**2022**
B.A. French	Chapman University	**2022**

- Earned the Ronald M. Huntington Outstanding Scholarship Award and DataFest First Place/Best Insight.

SKILLS

- Programming: Python, SQL, R, C/C++, JavaScript, MATLAB, Mathematica.
- Python Libraries: PyTorch, NumPy, pandas, Matplotlib.
- Machine Learning: Neural networks, SGD, logistic regression, gradient boosting, temporal validation, calibration.
- Quantitative Methods: Stochastic processes, optimization, Monte Carlo simulation, numerical analysis, kernel methods.
- Languages: English, French, Russian, Hebrew.

SELECTED PROJECTS

Deep Bellman Hedging and Portfolio Compression **2026 - Present**

- Implemented Python/PyTorch neural networks that choose stock holdings to offset option losses after trading costs; compared two ways of penalizing large losses (CVaR and entropic risk).
- Evaluated strategies on historical SPY data and Monte Carlo stock-price simulations (Black-Scholes), using separate training, validation, and test sets; compared with a standard hedge (delta hedging) and no hedge.

Complex Optimization and Kernel Regression **2026**

- Developed Python implementations of complex-valued gradient methods to reconstruct mathematical functions and study which solutions the algorithms learn.
- Established convergence guarantees under stated assumptions and checked them through numerical experiments.

Machine Learning for Drug and Vaccine Safety **2025 - 2026**

- Prioritized VAERS vaccine-symptom pairs for review using Python implementations of logistic regression and gradient boosting; evaluated next-quarter predictions against statistical signal scores. Presented at CHOC's Pediatric and Lifespan Data Science Conference (April 29, 2026).
- Refined Python gradient-boosting models to prioritize FAERS drug-safety signals, improving ranking performance (AUPRC) from approximately 0.68 to 0.85 on a pilot holdout of future quarters.

RESEARCH AND TEACHING EXPERIENCE

University of California, Irvine **Sep 2022 - Present**

Graduate Research and Teaching Assistant

- Investigate simpler portfolio summaries for managing risk through Deep Bellman Hedging, a current research direction.
- Proved tight bounds on how functions fluctuate across scales, with calculations implemented in Python; tested ideas suggested by large language models through proofs and computational checks.
- Taught Numerical Analysis I and II as instructor of record (Summer 2026), supported mathematics courses as a teaching assistant, and mentored DATA@UCI students in Python analysis and presentations (Jan-Jun 2025).
- Lead the Deep Hedging Directed Reading Program, guiding undergraduates through research papers, Python experiments, and critical evaluation.
- Mentored eight undergraduates on two random-matrix projects, guiding Python experiments and two research posters.

Chapman University **Jun 2020 - Sep 2022**

Machine Learning Research and Data Analysis

- Analyzed Brain Institute data using machine-learning models in Python and analysis tools in SQL and JavaScript.

SELECTED PUBLICATIONS AND PREPRINTS

Alpay & Battaglia. Complex Stochastic Gradient Descent and Directional Bias in Reproducing Kernel Hilbert Spaces. arXiv, 2026.
Alpay & Ivanisvili. Sharp Lower Bounds for Dyadic Square Functions of Indicator Functions of Sets. arXiv, 2025; revised 2026.
Alpay, Cerejeiras & Kähler. Generalized Fock-space and fractional derivatives: Uniqueness of sampling and interpolation sets. Results in Applied Mathematics, 2026.

Interests: Weightlifting, LEGO, karaoke.